
Sets and Constructions

How to write a collection down, and three ways to build a new one out of old ones.

Today

AGENDA

- 01 Two ways to write a set down
- 02 The sets you will use all semester
- 03 Subsets, equality, and Venn diagrams
- 04 Cardinality and power sets
- 05 Ordered tuples and Cartesian products



What Is a Set?

NOTATION

DEFINITION

A **set** is an unordered collection of objects, called the **elements** or **members** of the set. A set is said to **contain** its elements.

$x \in A$ — x is an element of A

$x \notin A$ — x is not

Unordered, and no repeats: $\{1, 2\}$, $\{2, 1\}$, and $\{1, 1, 2\}$ are all the same set.

The Roster Method

NOTATION

List all the members, when that is possible.

EXAMPLE

The set V of all vowels in the English alphabet:

$$V = \{a, e, i, o, u\}$$

Dots are allowed when the pattern is unmistakable: $\{1, 2, 3, \dots, 100\}$. They are not allowed when they are a guess.

Set-Builder Notation

NOTATION

Characterize the elements by stating the property they must have to be members.

THE VARIABLE

$\{ \text{x} \mid$

A stand-in for a typical element.

THE BAR

$|$

Reads “such that.” A colon means the same.

THE PROPERTY

$P(\text{x}) \}$

A predicate — exactly the thing from Lecture 2.

A set-builder expression is a predicate with braces around it. That is the whole idea.

Example: Odd Positives Under 10

NOTATION

ROSTER

$$O = \{1, 3, 5, 7, 9\}$$

SET-BUILDER

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

WITH A DOMAIN

$$O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$$

Naming the domain up front lets the property get shorter. It also removes an ambiguity — without $\mathbb{Z}^+$, is -3 in or out?

Sets You Will Use All Semester

NOTATION

$\mathbb{N}$	$\{0, 1, 2, \dots\}$	the natural numbers
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$\mathbb{Z}$	$\{\dots, -2, -1, 0, 1, 2, \dots\}$	the integers
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$\mathbb{Z}^+$	$\{1, 2, 3, \dots\}$	the positive integers
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$\mathbb{Q}$	$\{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$	the rationals
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$\mathbb{R}$	—	the reals
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Whether $0 \in \mathbb{N}$ is a convention, not a theorem. In this course it is in. Check every book you open.

Interval Notation

Interval	Set-builder
$[a, b]$	$\{x \mid a \leq x \leq b\}$
(a, b)	$\{x \mid a < x < b\}$
$[a, b)$	$\{x \mid a \leq x < b\}$
$(a, b]$	$\{x \mid a < x \leq b\}$

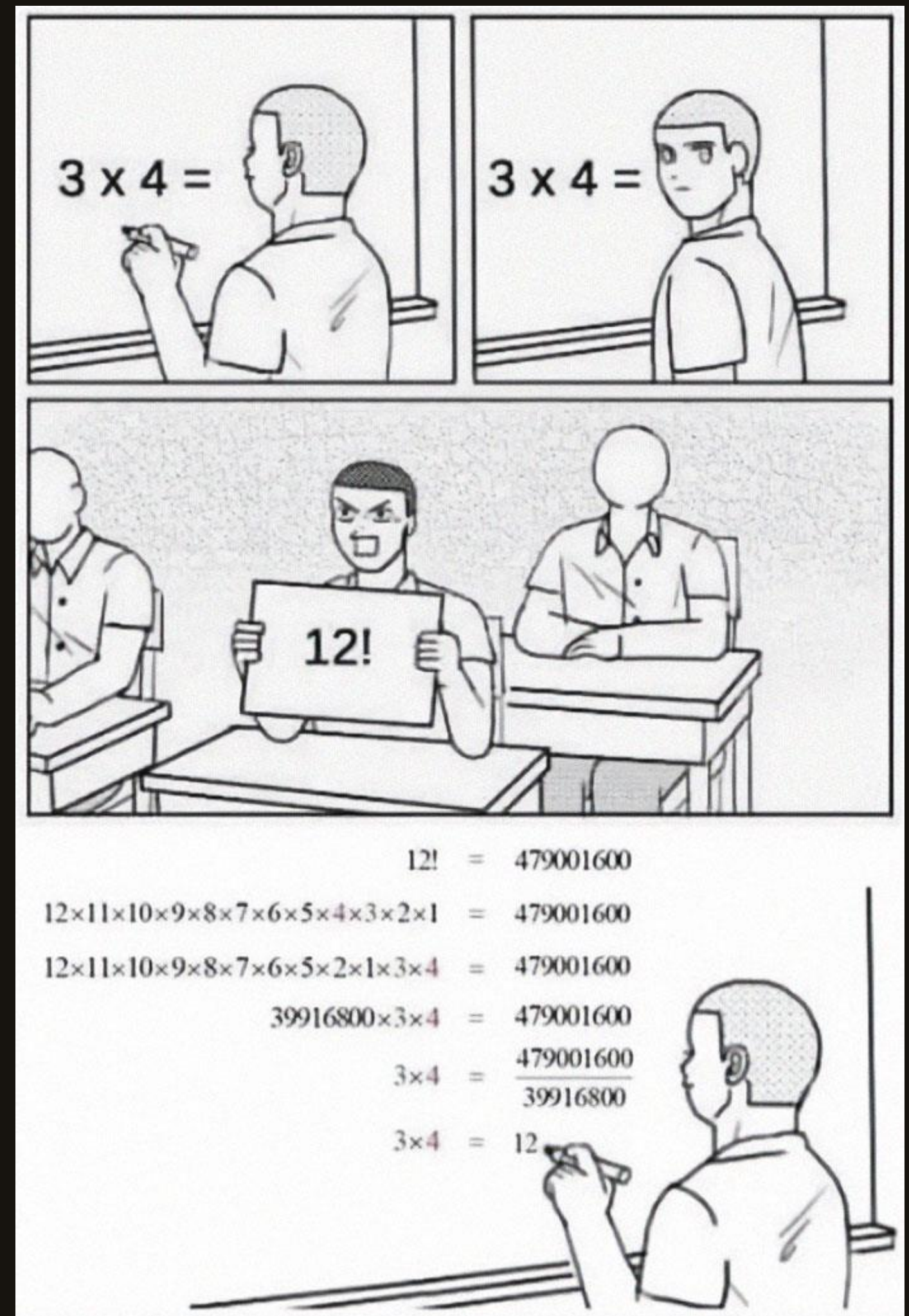
Square bracket: the endpoint is in. Round bracket: it is out.

These describe subsets of $\mathbb{R}$. There is no interval notation for the integers, because $\{2, 3, 4\}$ is easier to write than anything you could invent.

FOUR KINDS OF BRACKET

(a, b) is an interval. (a, b) is also an ordered pair.

Same two characters, two unrelated objects. Nobody is going to fix this either. Read the sentence around it.



The Empty Set

SUBSETS



The set with no elements. Also written $\{ \}$.

There is exactly one. Two sets with no elements have the same elements, so they are equal.

NOT THE SAME THING

$$|\emptyset| = 0$$

$$|\{\emptyset\}| = 1$$

An empty box, versus a box with an empty box in it.

Subsets

DEFINITION

The set A is a **subset** of B if and only if every element of A is also an element of B . We write $A \subseteq B$.

$$A \subseteq B \equiv \forall x (x \in A \rightarrow x \in B)$$

That is a universally quantified conditional — the exact shape from Lecture 2. Which tells you how every subset proof begins: *let x be an element of A .*

A Warning About $\subset$ and $\subseteq$

SUBSETS

SUBSET

$$A \subseteq B$$

Every element of A is in B . Equality allowed.

PROPER SUBSET

$$A \subsetneq B$$

Subset, and B has something A does not.

THE AMBIGUOUS ONE

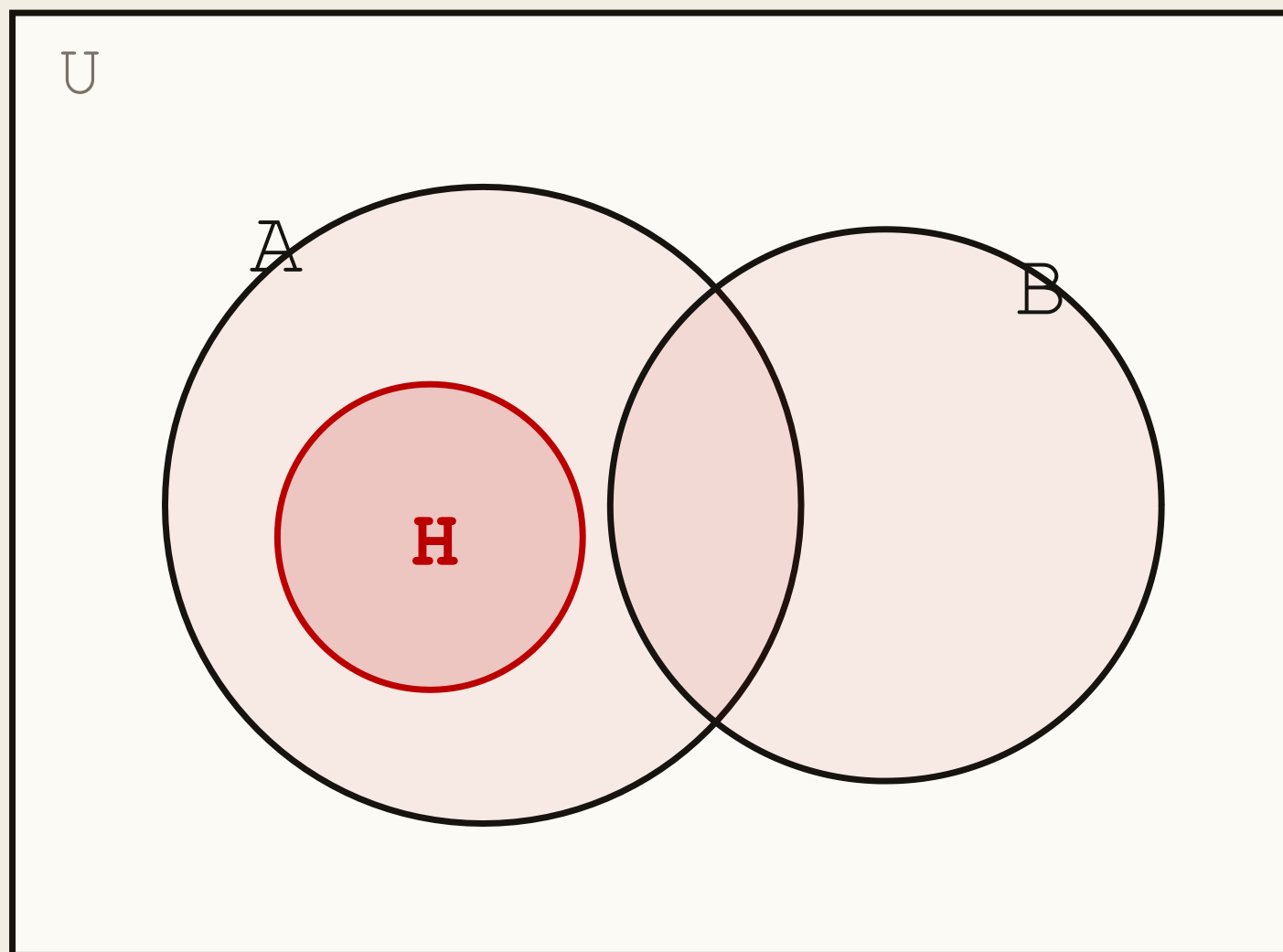
$$A \subset B$$

Means $\subseteq$ in some books and $\subsetneq$ in others.

It matters: $A \subseteq A$ is always true, and $A \subsetneq A$ is never true. In these slides $\subseteq$ always allows equality.

Venn Diagrams

SUBSETS



The rectangle is the **universal set** U — everything under consideration.

Circles inside it are sets. Here $H \subseteq A$, drawn as containment.

A diagram is a thinking aid. It shows you what to prove; it is not the proof.

Every Set Has Two Easy Subsets

SUBSETS

THEOREM

For every set A : $\emptyset \subseteq A$ $A \subseteq A$

WHY $\emptyset \subseteq A$

We need $\forall x \ (x \in \emptyset \rightarrow x \in A)$. The hypothesis $x \in \emptyset$ is false for every x , so every one of those implications is **vacuously true**. ■

WHY $A \subseteq A$

We need $\forall x \ (x \in A \rightarrow x \in A)$. Every implication of the form $p \rightarrow p$ is a tautology. ■

Vacuous truth, from the slide everybody argued about in Lecture 1, doing actual work.

Set Equality

CARDINALITY

DEFINITION

Two sets A and B are **equal** if $A \subseteq B$ and $B \subseteq A$.

$$A = B \equiv (A \subseteq B) \wedge (B \subseteq A)$$

Two containments, one equality. Compare the biconditional law from Lecture 1: $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.

This is *the* way to prove a set identity. Two paragraphs, one per direction.

Cardinality

DEFINITION

A set A with exactly n elements, n a nonnegative integer, is a **finite set**, and n is the **cardinality** of A , written $|A|$. A set is **infinite** if it is not finite.

$$|\{a, e, i, o, u\}| = 5$$

finite

$$|\emptyset| = 0$$

finite

$$\mathbb{Z}$$

infinite

Power Sets

DEFINITION

The **power set** of S is the set of all subsets of S ,
written $P(S)$.

$$P(S) = \{A \mid A \subseteq S\}$$

The elements of $P(S)$ are themselves sets.
That is the part to sit with.

By the theorem two slides back, $\emptyset$ and S are always
members.

Example: The Power Set of {1, 2, 3}

POWER SETS

$$P(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$$

SIZE 0

$\emptyset$

1 subset

SIZE 1

$\{1\}$ $\{2\}$ $\{3\}$

3 subsets

SIZE 2

$\{1,2\}$ $\{1,3\}$ $\{2,3\}$

3 subsets

SIZE 3

$\{1,2,3\}$

1 subset

$1 + 3 + 3 + 1 = 8$. Group by size and you will not miss one.

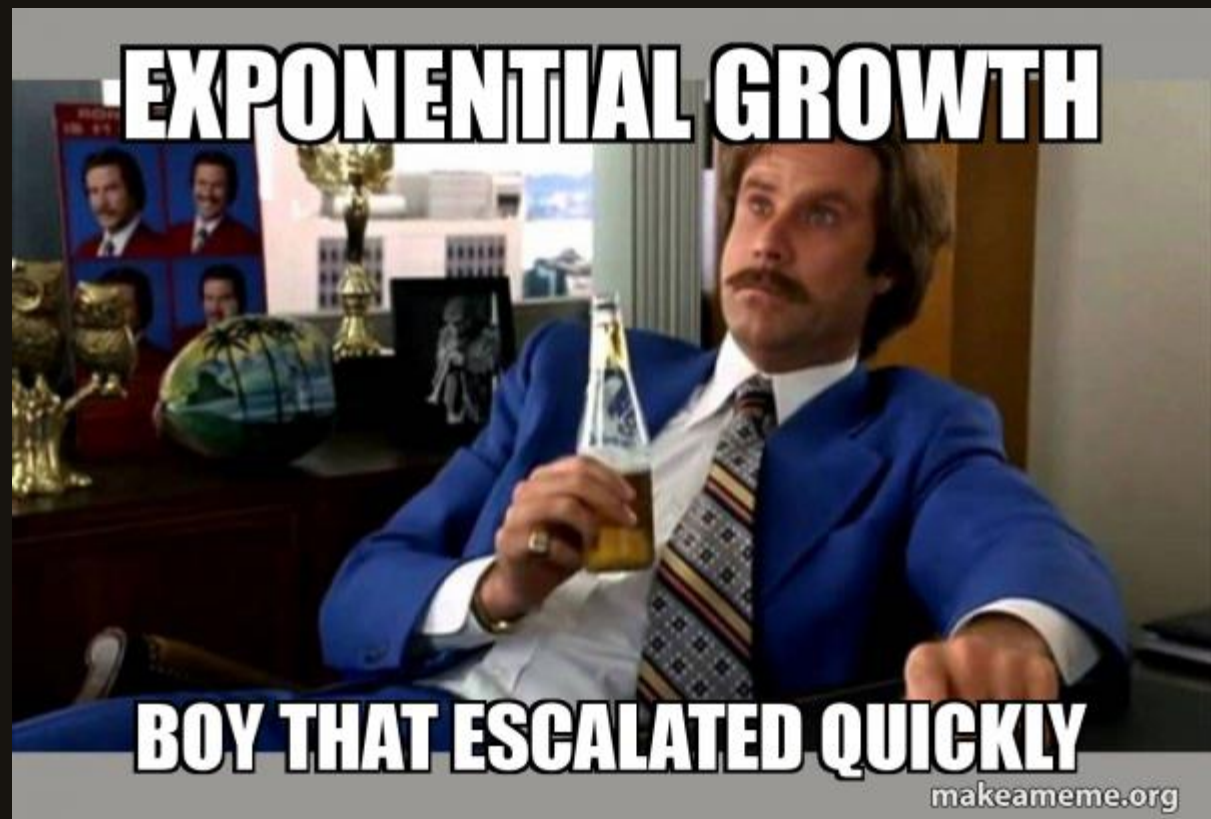
Why $|P(S)| = 2^n$

$$|S| = n \quad |P(S)| = 2^n$$

Building a subset means making one independent yes-or-no choice per element. Independent choices multiply.

So a subset of an n -element set *is* an n -bit string. That is not an analogy — it is how sets are stored.

Bits	Subset of $\{1, 2, 3\}$
000	$\emptyset$
001	$\{3\}$
010	$\{2\}$
011	$\{2, 3\}$
100	$\{1\}$
101	$\{1, 3\}$
110	$\{1, 2\}$
111	$\{1, 2, 3\}$



2^n , AGAIN

“List all subsets of a 20-element set.”

1,048,576 of them. The same curve as the truth tables in Lecture 2 — and not a coincidence: a truth assignment *is* a subset of the variables you made true.

Ordered n-Tuples

PRODUCTS

DEFINITION

The **ordered n-tuple** $(a_1, a_2, \dots, a_n)$ is the ordered collection with a_1 as its first element, a_2 as its second, and so on through a_n .

SETS

$$\{1, 2\} = \{2, 1\}$$

Unordered. Repeats collapse.

TUPLES

$$(1, 2) \neq (2, 1)$$

Ordered. Repeats are fine: $(1, 1)$ is a legitimate pair.

Two tuples are equal when they have the same length and agree position by position.

Cartesian Products

PRODUCTS

DEFINITION

The **Cartesian product** of A and B , written $A \times B$, is the set of all ordered pairs whose first element comes from A and whose second comes from B .

$$A \times B = \{ (a, b) \mid a \in A, b \in B \}$$

$$|A \times B| = |A| \cdot |B|$$

Order matters: $A \times B \neq B \times A$ unless the two sets are equal or one is empty.

Example: $A \times B$ as a Grid

PRODUCTS

$$A = \{1, 2, 3\} \quad B = \{a, b\}$$

$$A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$$

$$|A \times B| = 3 \cdot 2 = 6$$

	a	b
1	(1, a)	(1, b)
2	(2, a)	(2, b)
3	(3, a)	(3, b)

Common Mistakes

BEFORE YOU GO

- 01 Confusing $\in$ with $\subseteq$. For $S = \{1, 2\}$: $1 \in S$ is true, $1 \subseteq S$ is nonsense, and $\{1\} \subseteq S$ is true.
- 02 Treating $\emptyset$ and $\{\emptyset\}$ as the same set. One has no elements; the other has one.
- 03 Forgetting $\emptyset$ and S when listing $\mathcal{P}(S)$. They are always in there, and leaving them out breaks the count.
- 04 Writing $A \times B$ when you mean $B \times A$. The pairs are ordered, so the products differ.

Recap

- 01 A set is unordered and repeat-free. Write it by roster or by property.
- 02 $A \subseteq B$ is a quantified conditional, so subset proofs start “let $x \in A$.”
- 03 Equality is double containment. Prove both directions.
- 04 The power set collects every subset, and $|P(S)| = 2^n$ — one bit per element.
- 05 Cartesian products pair elements in order, and $|A \times B| = |A| \cdot |B|$.

Next: union, intersection, difference, complement — and a lot of Venn diagrams.